Climate policy analysis with endogenous preference change and innovation

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EPOC Conference

Outline

1. Introduction
2. The model
3. Calibration and validation
4. Simulation results

Motivation

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  - Aggregate outcomes caused by technological paradigms can change preferences.
- This interaction may have synergistic effects.

This project

Research questions

- What is the interaction effect between innovation and opinion dynamics in total emissions?
- How does the interaction term affect the optimal carbon tax?

Method

- Partial dis-equilibrium agent-based model with endogenous consumption preferences (OD model) and endogenous innovation (NK model).
- Case study applied to the automobile industry ($\sim$20%) of total emissions.

Previous contributions

- **Case studies**
  - (**Windrum2005**) finds that under heterogeneous preferences multiple stages of product innovation may occur, culminating in permanent design heterogeneity.
  - (**tripsas2008customer**) and (**windrum2009consumer**) finds that sudden changes in consumer preferences can cause new technological paradigms.
- **Simulation models**
  - (**adner2001demand**; **windrum2009consumer**; **herrmann2017optimal**) address how consumer preference heterogeneity alters the direction of innovations.
  - (**STUMMER2015157**) addresses how stochastic innovation shape the preferences of consumers in the Austrian Biofuel market.
  - (**roos2018endogenous**; **moore2022determinants**) proposes a system dynamics "earth-world" model where aggregate outcomes influence the opinion dynamics.
  - (**konc2021social**) finds a "social multiplier" that decreases the optimal carbon tax.

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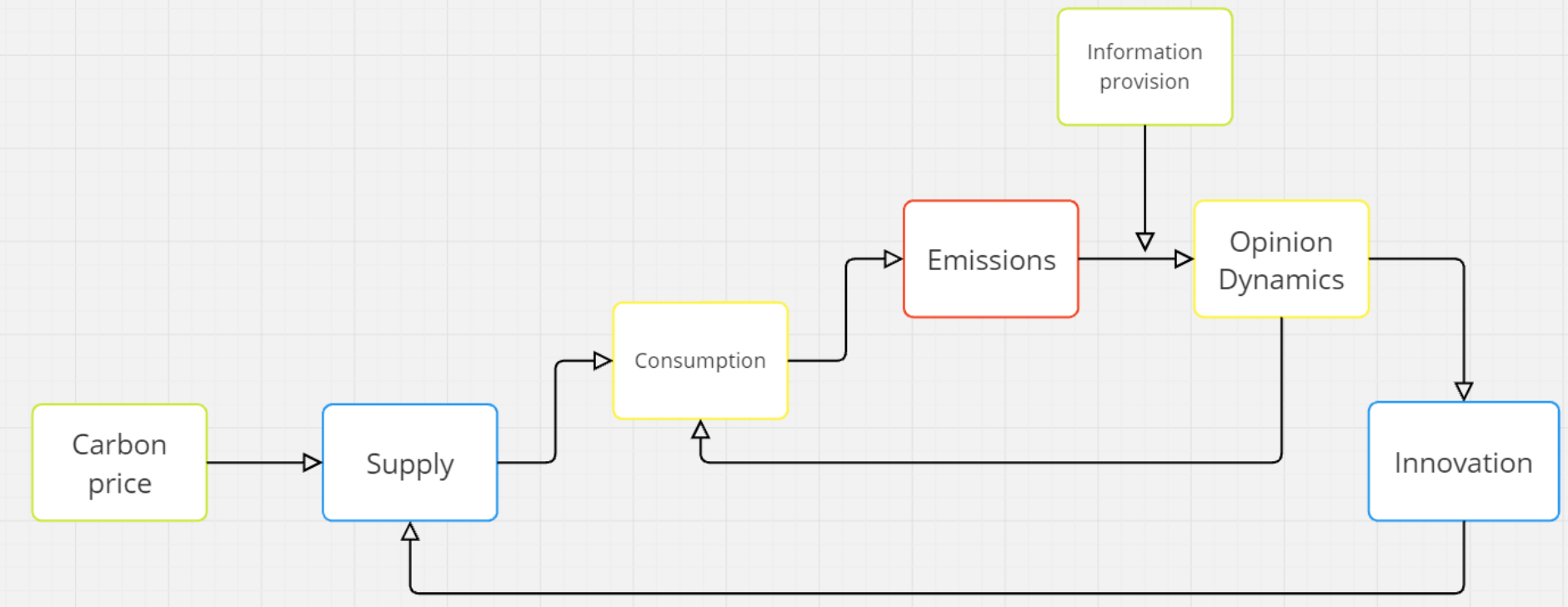
Model structure



Channels on the watch

Individual channels

- Innovation can reduce the overall emissions intensity of supplied goods, or increase the relative competitiveness of low-carbon goods.
- Opinion dynamics can increase the preferences for low-carbon goods and hence their consumption.

Interaction effects

- Changes in consumer preferences motivate changes in the direction of innovation, from cost reductions to emissions reductions.
- Innovation-caused increases in total emissions can increase climate awareness.

Consumption

1. Individuals observe the vehicles supplied Γ_t and evaluate their characteristics $\vec{X}_{g,t}$ (**Lancaster1979**; **McFadden1981**).
2. They consider buying a vehicle $g \in \Gamma_t$ with probability

$$\Phi(g' \in \Gamma_t, \Gamma_t, \vec{\beta}_{i,t}) = \frac{U^B_{i,t}(g')^{\kappa}}{\sum_{g=1}^{\Gamma_t} U^B_{i,t}(g)^{\kappa}} \tag{1}$$

   Where utility is a linear combination of price, quality and an environment score.
3. Buy vehicle if the utility is higher than the depreciated utility of the current car (facing no cost).

Opinion dynamics

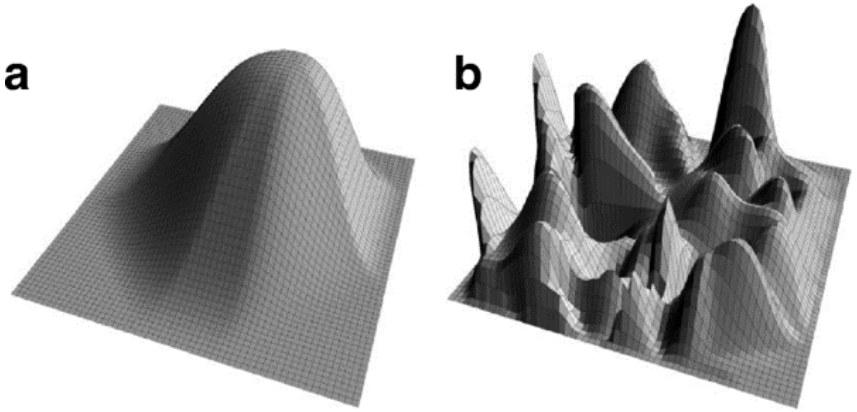
Individuals' opinion evolve with social influence and the perception of climate danger

$$\beta_{2,i,t} = (1-\upsilon)\beta_{2,i,t-1} + \upsilon\left((1-\upsilon_{E,i})S_{i,t} + \upsilon_{E,i}\frac{E_t}{E_{max}}\right) + \epsilon_t, \quad (2)$$

1. Current opinion
2. Social imitation
3. Emissions impact

Technologies: Fitness landscape

Firms locally search for technologies over an *NK* landscape (**kauffman1993origins**; **levinthal1997adaptation**).
Each landscape location return the product attributes (cost, quality, environment score).



Low K surface High K surface

Innovation: Search process

We use (**jain2014memory**) memory with neutrality algorithm.

1. Firms learn the population in market segments
2. Evaluate the expected profit of all neighbouring alternative technologies.
3. Classify technologies based on expected profit.
4. Acquire one technology of the same or higher rank

Evaluation function:

$$\Pi(v \in V, \Gamma \subset V, |\vec{I}_{w,t}|) = \mu X_v^C \sum_{w=1}^{W} |I_w| \Phi(v \in V, \Gamma \subset V, \vec{\beta}_w) \quad (3)$$

- Per unit profit.
- Population in market segment
- Expected in-segment market share

Supply

1. For each market segment $w : |I_w| \geq Y$, each firm $j \in J$ offers a car design.
2. Price equals costs plus an exogenous fixed mark-up.
3. **Carbon price**: Add $\tau(1 - X^C_{g,t})$ to the price.

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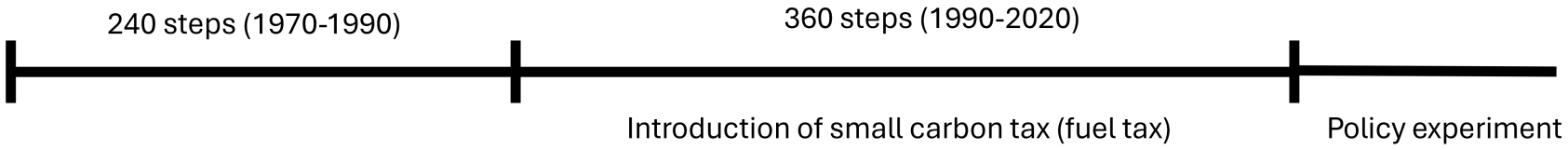
Calibration strategy

1. Parameters obtained from scientific articles: exogenous preferences (**ostli2017generic**), industry mark-up (**grieco2024evolution**), number of firms (**benkard2023concentration**), or carbon price (**aghion2016carbon**).
2. The non-observable parameters are calibrated indirectly to replicate documented trends of the automobile industry.
   - **T1**: Rise-and-fall trend in market concentration (**grieco2024evolution**).
   - **T2**: Reduction in costs and increasing quality followed by rising fuel efficiency and unchanged cost and quality (**aghion2016carbon**; **grieco2024evolution**).
   - **T3**: Persistent attribute heterogeneity among vehicles (**windrum2009consumer**).
   - **T4**: Increasing total emissions followed by a small decline in the last 4 years (OurWorldinData).



Validation: Market concentration

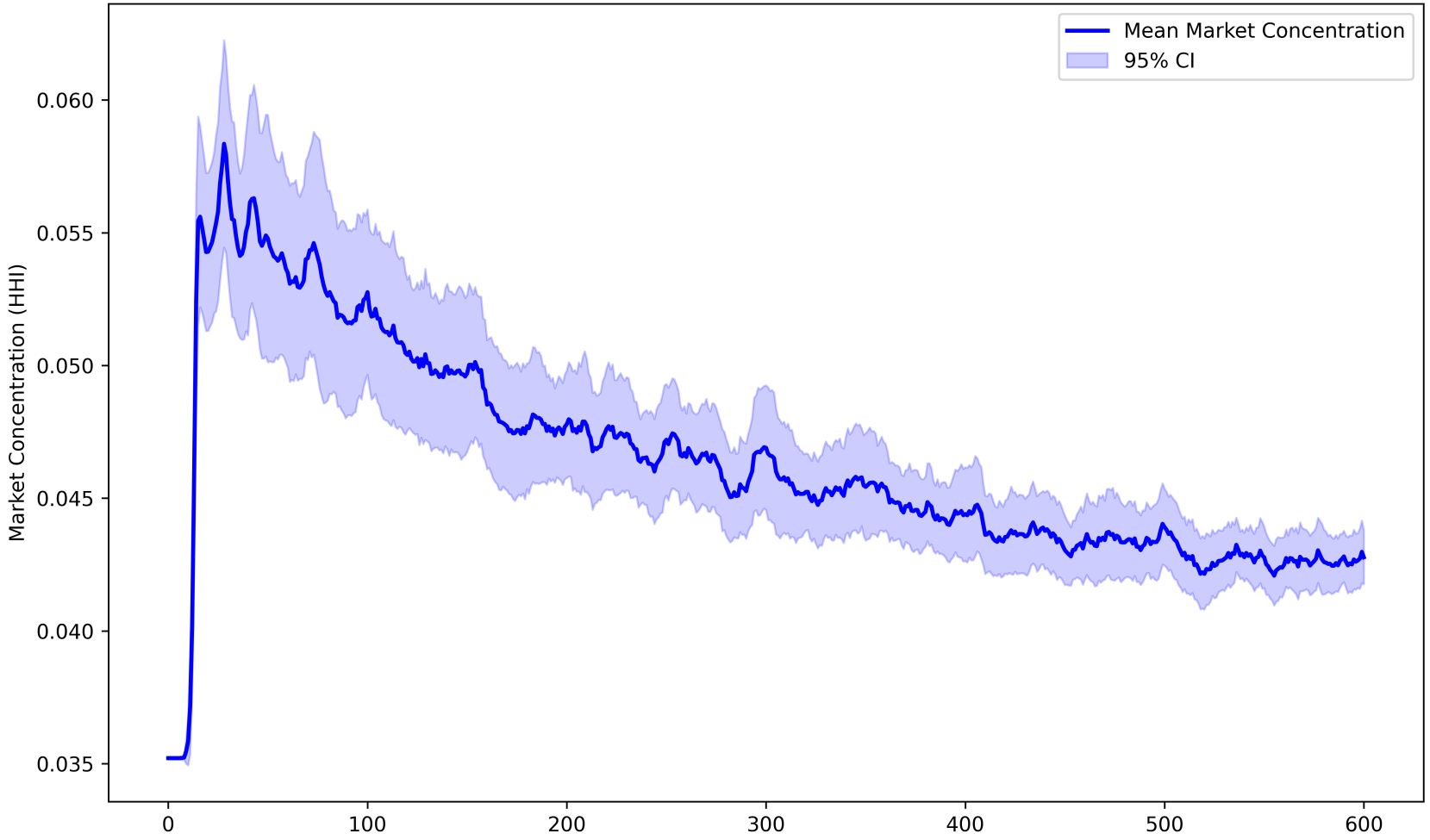


Figure: **T1**: Rise-and-fall trend in market concentration (**grieco2024evolution**)

Validation: Attributes

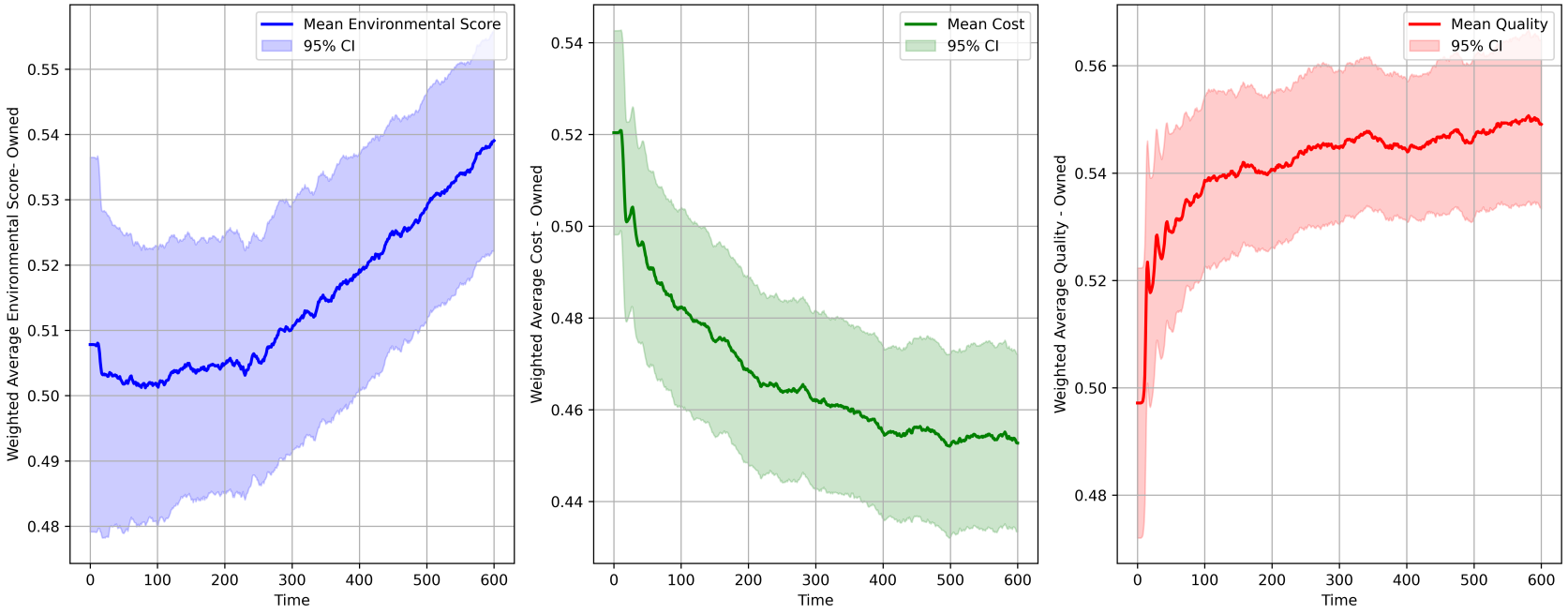


Figure: **T2**: Reduction in costs and increasing quality followed by rising fuel efficiency and unchanged cost and quality (**aghion2016carbon**; **grieco2024evolution**).

Validation: Heterogeneity

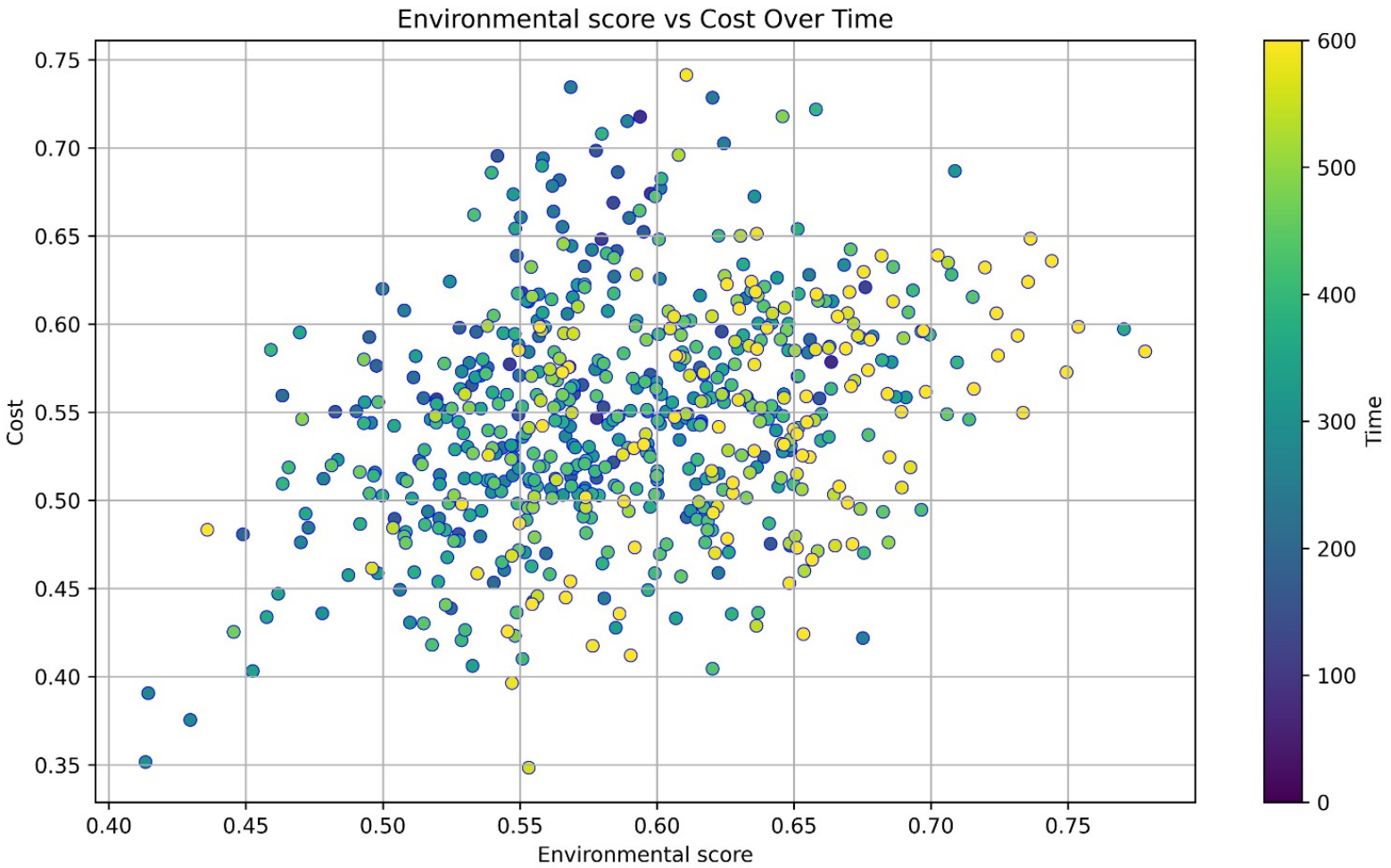


Figure: **T3**: Persistent attribute heterogeneity among vehicles (**windrum2009consumer**).

Validation: Emissions

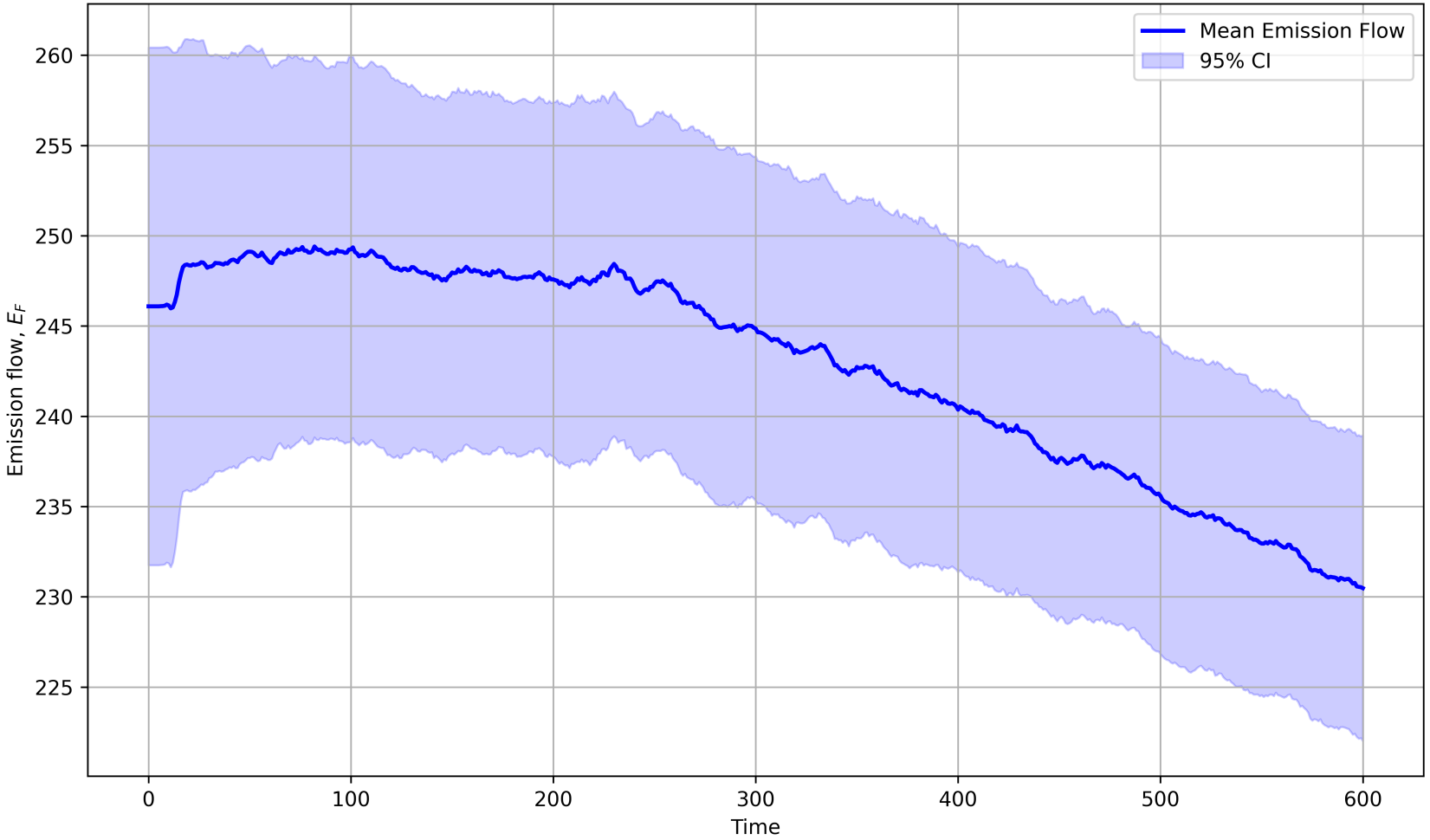


Figure: **T4**: Increasing total emissions followed by a small decline in the last 4 years (OurWorldinData).

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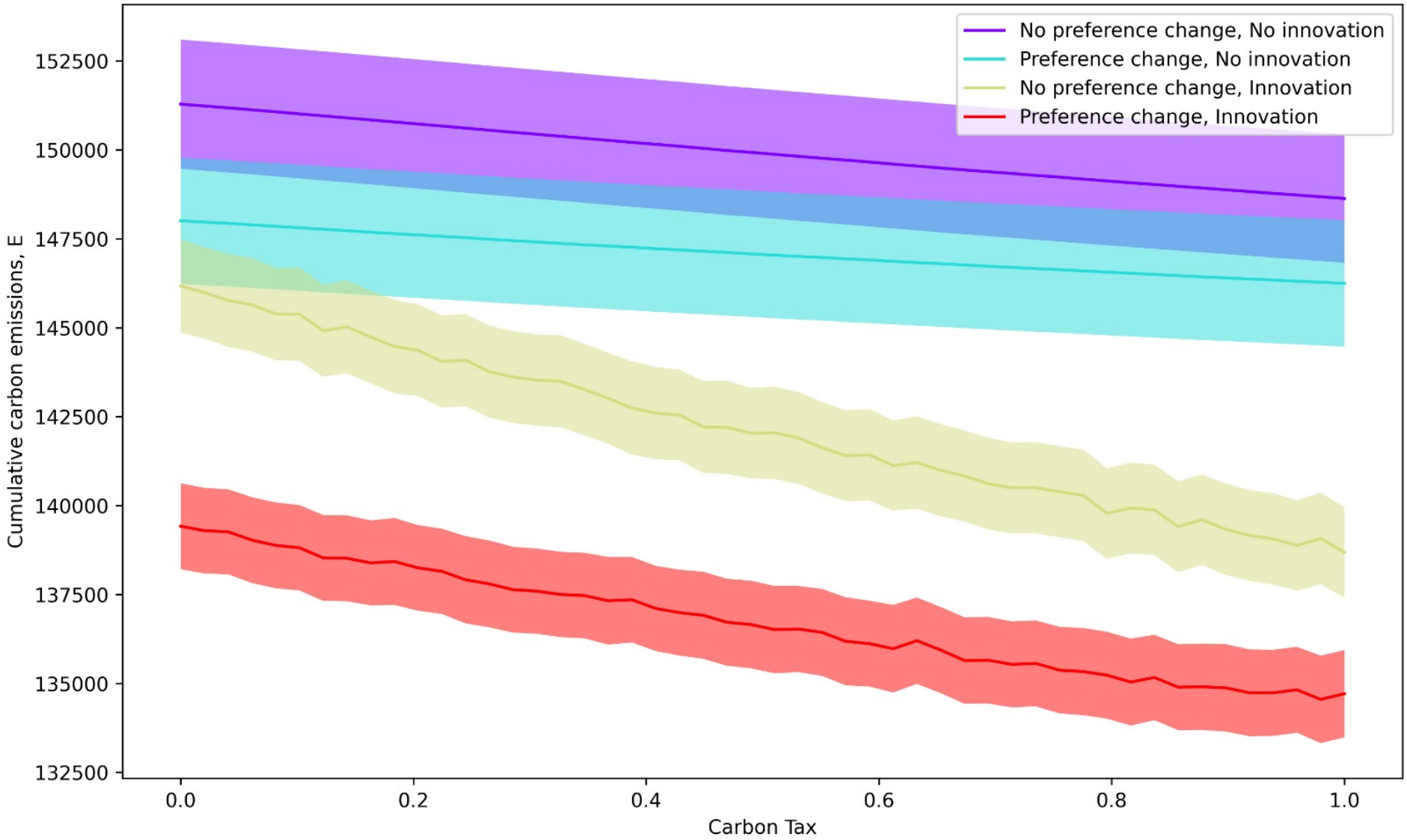
Approach

- RQ: What is the interaction effect between innovation and opinion dynamics?
- We simulate 400 runs for different carbon price values and four cases:
  1. No preference change and no innovation.
  2. Preference change and no innovation.
  3. No preference change and innovation.
  4. Preference change and no innovation.
- Perform a simple OLS regression to find the marginal effects on last time-step cumulative emissions.

Information provision: The experiments so far were inconclusive.

Results


No preference change, No innovation
Preference change, No innovation
No preference change, Innovation
Preference change, Innovation
Cumulative carbon emissions, E
152500
150000
147500
145000
142500
140000
137500
135000
132500
0.0
0.2
0.4
0.6
0.8
1.0
Carbon Tax

Results and intuitions

- There is an interaction effect only in levels: Changes in preferences speed innovation re-direction.
- Lower tax effect on emissions when considering opinion dynamics: substitute effects (concerned individuals are price-insensitive).
- When considering both channels, the carbon tax associated with an emissions target is always lower.

Concluding remarks

- We study the interaction between preference change and innovation in the context of the green transition in the automobile industry.
- We propose a partial dis-equilibrium agent-based model where both phenomena interact.
- We find a significant emission reduction interaction effect.
- Endogenous preferences act as a substitute for carbon taxes.
- Considering both channels simultaneously leads to a lower prescribed carbon tax.

Next steps

- Calibrate to validate the model quantitatively.
- Propose alternative social learning models with imperfect information.
- Allow for rebound effects with heterogeneous budgets.
- Experiment alternative policy experiments: information provision and public transportation.

Thank you!

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Bibliography I